In [16]:

```
from sklearn.feature_extraction.text import CountVectorizer
sents = ['He is a smart boy. She is also smart',
         'Chirag is a smart person']
```

In [18]:

```
cv = CountVectorizer()
X = cv.fit_transform(sents)
X = X.toarray()
vocabulary = sorted(cv.vocabulary_.keys())
print(vocabulary)
print(X)
```

```
['also', 'boy', 'chirag', 'he', 'is', 'person', 'she', 'smart']
[[1 1 0 1 2 0 1 2]
 [0 0 1 0 1 1 0 1]]
```

In [30]:

```
cv = CountVectorizer(ngram_range=(2,2))
X = cv.fit_transform(sents)
X = X.toarray()
vocabulary = sorted(cv.vocabulary_.keys())
print(vocabulary)
print(X)
```

```
['also smart', 'boy she', 'chirag is', 'he is', 'is also', 'is smart', 'she is', 'smart boy', 'smart person']
[[1 1 0 1 1 1 1 1 0]
 [0 0 1 0 0 1 0 0 1]]
```

In [49]:

```
from sklearn.feature_extraction.text import TfidfVectorizer
sents = ['coronavirus is a highly infectious disease',
   'coronavirus affects older people the most',
   'older people are at high risk due to this disease']
```

In [50]:

```
tfidf = TfidfVectorizer()
transformed = tfidf.fit_transform(sents)

import pandas as pd
df = pd.DataFrame(transformed[0].T.todense(),
                  index=tfidf.get_feature_names_out(),
                  columns=["TF-IDF"])
df = df.sort_values('TF-IDF', ascending=False)
df
```

Out[50]:

	TF-IDF
infectious	0.490479
highly	0.490479
is	0.490479
coronavirus	0.373022
disease	0.373022
older	0.000000
this	0.000000
the	0.000000
risk	0.000000
people	0.000000
affects	0.000000
most	0.000000
are	0.000000
high	0.000000
due	0.000000
at	0.000000
to	0.000000

In [9]:

```
import re
from nltk.util import ngrams
from nltk.tokenize import word_tokenize

text = """Natural language processing refers to the branch of
          computer science concerned with giving computers
          the ability to understand text"""
words = word_tokenize(text)
output = list(ngrams(words, 2))
output
```

Out[9]:

```
[('Natural', 'language'),
 ('language', 'processing'),
 ('processing', 'refers'),
 ('refers', 'to'),
 ('to', 'the'),
 ('the', 'branch'),
 ('branch', 'of'),
 ('of', 'computer'),
 ('computer', 'science'),
 ('science', 'concerned'),
 ('concerned', 'with'),
 ('with', 'giving'),
 ('giving', 'computers'),
 ('computers', 'the'),
 ('the', 'ability'),
 ('ability', 'to'),
 ('to', 'understand'),
 ('understand', 'text')]
```

In [ ]:

```

```